



TOTAL MARKS: 40
DATE: 08/07/24

Question 1 **[1x10=10]**

- 1

- vi) **Assertion (A):** AI systems can improve decision-making processes in critical areas such as healthcare.
Reason (R): AI systems can process large amounts of data quickly and identify patterns that humans might miss.
- Both A and R are true, and R is the correct explanation of A.
 - Both A and R are true, but R is not the correct explanation of A.
 - A is true, but R is false.
 - A is false, but R is true.
- vii) Identify the diagram of ultrasonic sensor from the given diagram-
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- viii) What is one of the primary **ethical** concerns related to the use of AI technology?
- Decreased efficiency in tasks
 - Potential for bias and discrimination
 - Limited scalability
 - High cost of implementation
- ix) Deep learning includes algorithms based on the concept of artificial neural network. (True or False)
- x) The light sensor detects the presence of an object. (True or False)

Question 2

- i) The following Python code prints a **hollow square**. Some part of the code is missing. Fill in the blanks so that it meets the objective. After substitution study the code and answer the following questions-

```
size = 5
for i in range(size):
    for j in range(_?1?_):
        if i == __?2?__ or i == size - 1 or j == __?3?__ or j == size - 1:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

- a) What is the expression/ value at ?1? [1]
 - b) What is the expression/ value at ?2? [1]
 - c) What is the expression/ value at ?3? [1]
 - d) What is the role of **end** in print statement? [1]
 - e) Draw the output. [1]
- ii) The following program is checking if the given number is a **prime**. The code is erroneous, hence, not giving the right output. Correct the errors and re-write the code. Underline the corrections. [5]

```

define isPrime(n):
    if n < 1:
        return False
    for i in range(2, n):
        if n % i == 0:
            return False;
    return true

```

Question 3

- i) Write a program in Python to generate **Fibonacci** series upto '**n**' terms. [7]
- Input:
N= 8
- Output:
1 1 2 3 5 8 13 21
- ii) What is the purpose of **actuators** in robots? Name any two types of actuators. [3]

Question 4

- i) Define a method in Python that checks if '**n**' is a **palindrome** number. A *palindrome number is a number that reads the same forward and backward*. Example 3, 5, 121, 23432 etc [7]
- ii) Discuss some **ethical** issues in terms of AI. [3]